



Mk 19 grenade launcher

The **Mk 19 grenade launcher** (pronounced Mark 19) is an American 40 mm belt-fed automatic grenade launcher that was first developed during the Vietnam War.^{[7][8]}

Overview



A U.S. Marine fires a Mk 19 40 mm grenade launcher in Quantico, Virginia in September 2000.

The Mk 19 is a belt-fed, blowback-operated, air-cooled, crew-served, fully-automatic weapon that is designed not to cook off. It fires 40 mm grenades at a cyclic rate of 325 to 375 rounds per minute, giving a practical rate of fire of 60 rounds per minute (rapid) and 40 rounds per minute (sustained). The weapon operates on the blowback principle, which uses the chamber pressure from each fired round to load and re-cock the weapon. The Mk 19 can launch its grenade at a maximum distance of 2,212 meters (2,419 yd), though its effective range to a point target is about 1,500 meters (1,600 yd), since the large rear leaf sight is only graduated as far. The nearest safe distance to launch the grenade is 310 meters in training and 75 meters in combat. Though the Mk 19 has a flash suppressor, it serves only to save the eyesight of its operator, not concealing the

Mk 19



A Mk 19 40 mm grenade launcher mounted on an M3 tripod

Type	<u>Automatic grenade launcher</u>
Place of origin	<u>United States</u>

Service history

In service	1968–present
Used by	See <u>Users</u>
Wars	<u>Cold War</u> <u>Vietnam War</u> <u>Sino-Vietnamese War</u> <u>Cambodian–Vietnamese War</u> <u>Sino-Vietnamese conflicts (1979–1991)</u> <u>Third Indochina War</u> <u>Soviet–Afghan War</u> <u>Persian Gulf War</u>^[1] <u>Global War on Terrorism</u> <u>War in Afghanistan</u> <u>Iraq War</u> <u>2006 Lebanon War</u> <u>Kurdish–Turkish conflict (1978–present)</u>^{[2][3]} <u>Mexican Drug War</u> <u>2013 Lahad Datu standoff</u> <u>Syrian Civil War</u>^[4] <u>Yemeni Civil War</u> <u>Saudi Arabian-led intervention in Yemen</u>

weapon's position. For night operation, a picatinny rail quadrant sight can be added for thermal and night vision optics.

The Mk 19A is a man-portable crew-served weapon that can fire from a tripod-mounted position or from a vehicle mount, with the latter being the preferred method, as the weapon alone weighs 77.6 pounds (35.2 kg). The primary ammunition for it is the high-explosive dual-purpose M430 grenade. On impact, the grenade can kill anyone within a radius of five metres (16 ft), and wound them within a radius of 15 metres (49 ft). It can also penetrate two inches (5.1 cm) of rolled homogeneous armor with a direct hit (0-degree obliquity), which means it can penetrate most infantry fighting vehicles and armored personnel carriers. It is especially effective when used against enemy infantry formations. The ammunition comes in cans that hold a 32- or 48-grenade belt weighing 42 and 60 pounds (19 and 27 kg), respectively. Due to its low recoil and comparatively light weight, it has been adapted for use on many different platforms, including small attack boats, fast attack vehicles such as the Humvee (HMMWV), AAV and Stryker, military jeeps, and a large variety of naval mounts.

The Mk 19 automatic grenade launcher replaced the earlier Mk 18 hand-cranked multiple grenade launcher. The 40 mm ammunition used (40×53 mm) is not interchangeable with that used in the M203 (40×46 mm). The M203 ammunition develops a lower chamber pressure, and resultant lower muzzle velocity and range, compared to ammunition loaded for the Mk 19. The Mk 19 fires from an open bolt. The rounds are mechanically fed onto the bolt face with the pull of the charging handles. When the trigger is pressed, the bolt closes, and the firing pin is released. The recoil blows back the bolt, feeds a new round onto the bolt face, which pushes the expended casing off the bolt face.

Production of the Mk 19 is managed by Saco Defense Industries (now a division of General Dynamics Ordnance and Tactical Systems).

Russo-Ukrainian War Gaza war	
Production history	
Designer	Naval Ordnance Station Louisville, George M. Chinn
Designed	1966
Manufacturer	Saco Defense Industries (now a division of <u>General Dynamics Ordnance and Tactical Systems</u>), Combined Service Forces, <u>MKEK</u> ^[5]
Unit cost	\$20,000 ^[6]
Produced	1968–present
Variants	Mk 19 Mod 0, Mk 19 Mod 1, Mk 19 Mod 2, Mk 19 Mod 3, Mk 19 Mod 4
Specifications	
Mass	77.6 pounds (35.2 kg) (empty, without accessories)
Length	43.1 inches (1,090 mm)
Barrel length	16.25 inches (413 mm)
Width	9.46 inches (240.4 mm)
Height	7.8 inches (199 mm)
Cartridge	40×53 mm
Action	API blowback
Rate of fire	325-375rpm (cyclic) 40 rpm (sustained) 60 rpm (rapid)
Muzzle velocity	750–790 feet per second (230–240 m/s) (average)
Effective firing range	1,500 m (1,600 yd)
Maximum firing range	2,212 m (2,419 yd)
Feed system	32- or 48-grenade belt
Sights	Iron

In November 2014, General Dynamics entered into an agreement with Advanced Material Engineering Pte Ltd, a subsidiary of Singapore Technologies Kinetics, to manufacture 40 mm high-velocity airburst ammunition for the U.S. military. The 40 mm airburst grenade uses a programmable, time-based fuse that computes and programs the detonation time into it, which counts down once fired to zero to detonate at the intended target point. The airburst ammunition is compatible with the Mk 19, which would give it greater effectiveness and lethality, particularly against concealed and defilade targets.

The Mk 19 is used as part of the Şahin counter unmanned air system (C-UAS) developed by Aselsan, which entered service with the Turkish Armed Forces in July 2022 and uses Atom 40 mm ammunition to detect and destroy mini and micro UAVs out to 700 meters.^{[9][10]}

Variants

"Although the MK 19 is a recent entry into the Army's inventory, development began in 1963. The first version was a hand-cranked, multiple grenade launcher called the MK 18. In 1966 the need for more firepower inspired the development of a self-powered 40-mm machine gun called the MK 19, MOD 0. This model was neither reliable nor safe enough for use as a military weapon system. Product improvements begun in 1971 resulted in the 1972 MOD 1, of which only six were produced. The MOD 1 performed effectively in Navy riverine patrol craft and broader applications for the MK 19 were found. In 1973 the Navy developed the MOD 2, which featured improved reliability, safety, and maintainability. In 1976 a complete redesign resulted in the MK 19, MOD 3, which the Army adopted in 1983. The Army now uses the MK 19 within the tactical environment for defense, retrograde, patrolling, rear area security, urban operations, and special operations."^[11] - Department of the Army. November 2003 -

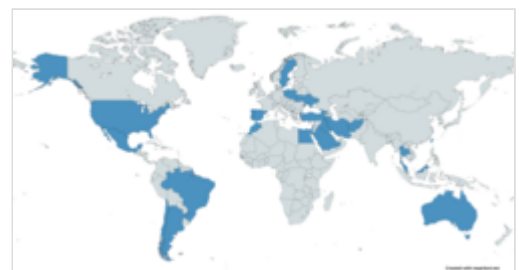


US Navy crewman fires a Mk 19 Mod 0 in July 1969

Users

GDOTS has built nearly 35,000 Mk 19 Mod 3 systems for roughly 30 customers since 1984.^[12] Users of the Mk 19 include:

-  Afghanistan: Unknown numbers were given to the Afghan army at an unknown period of time, captured and in use by the Taliban.^[13]
-  Argentina: Argentine Marines.^[14]
-  Australia^[15]
-  Brazil: Used by the Brazilian Marine Corps.^[16]
-  Chile: Used by the Chilean Army and the Chilean Marine Corps.^{[17][18]}
-  Egypt: Manufactured locally by Helwan.^{[19][20]}



Map with Mk 19 grenade launcher users in blue

-  Georgia^[21]
-  Iran^[22]
-  Iraq: Used by Iraqi special forces on Humvees.^[23]
-  Israel:^[15] Adopted by the Israeli Defence Forces (under the name "Maklar", for *mikla rimonim* or "grenade machinegun"), to be fielded in infantry and mechanized units. The Mk 19 was formerly manufactured locally.^[19]
-  Japan: MK19 mounted on AAV7 amphibious vehicle used by Amphibious Rapid Deployment Brigade^[24]
-  Kosovo:^[15] Kosovo Security Forces.^[19]
-  Lebanon^[25]
-  Malaysia^[15]
-  Mexico:^[15]
-  Moldova^{[26][27]}
-  Morocco: Moroccan Army.^[28]
-  Poland^[29]
-  Portugal: Used by the Portuguese Army.^[30]
-  Saudi Arabia^[31]
-  Spain^[15]
-  Sweden: Designated *Grsp 92*.^[15] Used by Kustjägarna and Amfibiebataljonen^[32] and also by the 31st Airborne Battalion^[33]
-  Taiwan^[1]
-  Thailand:^[34]
-  Turkey: Produced under licence by MKEK.^[5] Used by Turkish Land Forces.^[35]
-  Ukraine^[36]
-  United States:^[15] Currently in widespread use throughout the U.S. Armed Forces.
-  Vietnam:^[15] Used in the Sino-Vietnamese War.



U.S. Navy sailors fire a Mk 19 40 mm grenade launcher during a training exercise in March 2003.



Mk 19 in use by Polish Land Forces



A Mexican Army Chevrolet Silverado equipped with a Mk 19 at a military checkpoint in March 2009

See also

- Mk 47 Mod 0 Striker, U.S. military successor in limited service
- XM174 grenade launcher, predecessor used by the U.S. military during the Vietnam War
- U.S. Navy equipment designations
- List of API blowback firearms
- List of grenade launchers
- Comparison of automatic grenade launchers

Weapons of comparable role, performance and era

- AGS-17 – (Soviet Union)
- AGS-30 – (Russia)
- AGS-40 – (Russia)
- Heckler & Koch GMG – (Germany)
- QLZ-87 – (China)
- QLZ-04 – (China)
- K4 grenade launcher – (South Korea)
- Howa Type 96 – (Japan)

- UAG-40 – (Ukraine)

References

1. "Report: Profiling the Small Arms Industry - World Policy Institute - Research Project" (<http://web.archive.org/web/20171011093831/http://www.worldpolicy.org/projects/arms/reports/smallarms.htm>). World Policy Institute. November 2000. Archived from the original (<http://www.worldpolicy.org/projects/arms/reports/smallarms.htm>) on 2017-10-11. Retrieved 2010-07-15.
2. #Devon (12 August 2017). "kurds attack mountain top Turkish positions pt 2 Daring close quarters battle against regime" (<https://www.youtube.com/watch?v=UEH5ihLou7M>) – via YouTube.
3. "Özel birlikler Kato'yu PKK'ya dar etti (Special troops deny the Kato [Mountain] to the PKK)" (<http://gundem.milliyet.com.tr/ozel-birlikler-kato-yu-pkk-ya-dar-etti/gundem/gundemdetay/10.09.2012/1594102/default.htm>). *Milliyet*. 2012-09-11. Retrieved 2012-11-24.
4. Kênh Tổng Hợp (27 February 2017). "CHIẾN SỰ MOSUL. ĐỈNH ĐIỂM KHỐC LIỆT 02 2017" (<https://www.youtube.com/watch?v=cc2wN-MIsfs>) – via YouTube.
5. "MKEK - Makina ve Kimya Endüstrisi Kurumu" (<https://web.archive.org/web/20160306070932/http://www.mkek.gov.tr/tr/Urunler.aspx?UretimYeriID=0&UrunID=301&AnaKategori=107&AltKategori=122&EnAltKategori=>). *mkek.gov.tr*. Archived from the original (<http://www.mkek.gov.tr/tr/Urunler.aspx?UretimYeriID=0&UrunID=301&AnaKategori=107&AltKategori=122&EnAltKategori=>) on 6 March 2016. Retrieved 4 November 2016.
6. "Weapons: Mk19 Gets A Brain" (<http://www.strategypage.com/htmwh/weap/articles/20110330.aspx>). Strategypage.com. 2011-03-30. Archived (<https://web.archive.org/web/20121012190944/http://www.strategypage.com/htmwh/weap/articles/20110330.aspx>) from the original on 2012-10-12. Retrieved 2012-11-24.
7. "Field Manual 3-22.27 (FM 23.27) MK 19, 40-mm GRENADE MACHINE GUN, MOD 3" (http://armypubs.army.mil/doctrine/DR_pubs/dr_a/pdf/fm3_22x27.pdf) (PDF). Department of the Army. November 2003. Archived (https://web.archive.org/web/20160304055631/http://armypubs.army.mil/doctrine/DR_pubs/dr_a/pdf/fm3_22x27.pdf) (PDF) from the original on 2016-03-04. Retrieved 2015-06-05.
8. Tucker, Spencer C. (20 May 2011). *The Encyclopedia of the Vietnam War: A Political, Social, and Military History [4 volumes]: A Political, Social, and Military History* (<https://books.google.com/books?id=qh5lffww-KsC&pg=PA431>). ABC-CLIO. p. 431. ISBN 978-1-85109-961-0. Archived (<https://web.archive.org/web/20160517145920/https://books.google.com/books?id=qh5lffww-KsC&pg=PA431>) from the original on 17 May 2016. Retrieved 3 November 2016.
9. "Turkey receives first Sahin counter-UAV system" (<https://www.janes.com/defence-news/news-detail/turkey-receives-first-sahin-counter-uav-system#:~:text=Aselsan%27s%20%C5%9Eahin%20counter%20unmanned%20aerial,%20C%20tweeted%20on%2017%20July.>). *Janes Information Services*. 19 July 2022. Archived (<https://web.archive.org/web/20220724215907/https://www.janes.com/defence-news/news-detail/turkey-receives-first-sahin-counter-uav-system>) from the original on 24 July 2022.
10. "ŞAHİN 40MM FİZİKSEL İMHA SİSTEMİ" (https://www.aselsan.com.tr/SAHIN40mmTR_1426.pdf) (PDF). *Aselsan.com.tr*. Retrieved 27 July 2022.
11. "Field Manual 3-22.27 (FM 23.27) MK 19, 40-mm GRENADE MACHINE GUN, MOD 3" (http://armypubs.army.mil/doctrine/DR_pubs/dr_a/pdf/fm3_22x27.pdf) (PDF). Department of the Army. November 2003. Archived (https://web.archive.org/web/20160304055631/http://armypubs.army.mil/doctrine/DR_pubs/dr_a/pdf/fm3_22x27.pdf) (PDF) from the original on 2016-03-04. Retrieved 2015-06-05.

12. Ian Kemp. "Heavy Hitters : Heavy Machine Guns and Grenade Launchers" (<https://web.archive.org/web/20110707174856/http://www.asianmilitaryreview.com/upload/200711272151281.pdf>) (PDF). Archived from the original (<http://www.asianmilitaryreview.com/upload/200711272151281.pdf>) (PDF) on 2011-07-07. Retrieved 27 July 2022.
13. Hilal Ghaznawi (2022-09-11). "Mk 19 Grenade Launcher in Taliban hands" (https://web.archive.org/web/20220914213100/https://twitter.com/ghaznawi_hilal/status/1569048016988041217). *Twitter*. Archived from the original (https://twitter.com/ghaznawi_hilal/status/1569048016988041217) on 2022-09-14. Retrieved 2022-09-11.
14. "Armada Argentina - official site" (https://web.archive.org/web/20071012033447/http://www.ara.mil.ar/pnav_infmarina.asp#02). Estado Mayor General de la Armada. Archived from the original (http://www.ara.mil.ar/pnav_infmarina.asp#02) on 2007-10-12.
15. Jones, Richard D. *Jane's Infantry Weapons 2009/2010*. Jane's Information Group; 35 edition (January 27, 2009). ISBN 978-0-7106-2869-5.
16. "Corpo de Fuzileiros Navais adquire 23 novos CLAnfs - Noticias Infodefensa América" (<http://www.infodefensa.com/latam/2014/04/02/noticia-corpo-fuzileiros-navais-adquire-novos-clanfs.html>). *infodefensa.com*. 2 April 2014. Archived (<https://web.archive.org/web/20161008194329/http://www.infodefensa.com/latam/2014/04/02/noticia-corpo-fuzileiros-navais-adquire-novos-clanfs.html>) from the original on 8 October 2016. Retrieved 4 November 2016.
17. García, Nicolás (20 July 2021). "The Chilean Army performs combat shooting with Saab's Carl Gustaf anti-armor system" (<https://www.infodefensa.com/texto-diario/mostrar/3110950/ejercito-chile-realiza-tiro-combate-sistema-antiblandaje-carl-gustaf-saab>). *Infodefensa.com*. Retrieved 22 July 2022.
18. García, Nicolás (1 July 2017). "The Chilean Marine Infantry exhibits its most modern rifles and machine guns in Concón" (<https://www.infodefensa.com/texto-diario/mostrar/3077421/infanteria-marina-chile-exhibe-concon-modernos-fusiles-ametralladoras>). *Infodefensa.com*. Retrieved 22 July 2022.
19. "A new generation of AGLs" (<http://www.thefreelibrary.com/A+new+generation+of+AGLs%3A+within+only+a+few+decades+the+Automatic...-a086195789>). Thefreelibrary.com. 2002-04-01. Archived (<https://web.archive.org/web/20110430050116/http://www.thefreelibrary.com/A+new+generation+of+AGLs%3A+within+only+a+few+decades+the+Automatic...-a086195789>) from the original on 2011-04-30. Retrieved 2012-11-24.
20. "IDEX ABU 2017 – Small Arms Defense Journal" (<http://www.sadefensejournal.com/wp/?p=4132>). Archived (<https://web.archive.org/web/20180123072923/http://www.sadefensejournal.com/wp/?p=4132>) from the original on 2018-01-23. Retrieved 2018-09-30.
21. "Georgia Receives New Military Weaponry with U.S. Support" (<https://civil.ge/archives/396867>). *Civil.ge*. 11 February 2021.
22. "Great Prophet 5 Maneuvers (5)" (<http://www.uskowioniran.com/2010/04/great-prophet-5-naval-maneuvers-5.html?m=1>). *uskowioniran.com*. Archived (<https://web.archive.org/web/20170202002057/http://www.uskowioniran.com/2010/04/great-prophet-5-naval-maneuvers-5.html?m=1>) from the original on 2 February 2017. Retrieved 4 November 2016.
23. Vining, Miles (June 19, 2018). "ISOF Arms & Equipment Part 4 – Grenade Launchers & Anti-Armour Weapons" (<http://armamentresearch.com/isof-arms-equipment-part-4-grenade-launchers-anti-armour-weapons/>). *armamentresearch.com*. Archived (<https://web.archive.org/web/20180728192622/http://armamentresearch.com/isof-arms-equipment-part-4-grenade-launchers-anti-armour-weapons/>) from the original on July 28, 2018. Retrieved July 31, 2018.
24. "水陸両用車AAV7" (<https://rikuzi-chousadan.com/soubihin/soukousya/aav7.html>) (in Japanese). Retrieved Oct 23, 2022.

25. Miles, Donna (April 8, 2009). "Gates, Lebanese Defense Minister Explore Expanding Bilateral Relationship" (<http://www.defenselink.mil/news/newsarticle.aspx?id=53846>). American Forces Press Service - DefenseLink News. Archived (<https://web.archive.org/web/20090411030408/http://www.defenselink.mil/news/newsarticle.aspx?id=53846>) from the original on April 11, 2009. Retrieved 2009-04-08.
26. Cristina Mirca (4 August 2023). "SUA au donat Armatei Naționale a R. Moldova un nou lot de echipamente militare, în valoare de trei milioane de dolari, în urma unui acord între cele două țări, semnat în anul 2019" (<https://tvrmdova.md/article/ffbfad4a25b1889d/sua-au-donat-armatei-nationale-a-r-moldova-un-nou-lot-de-echipamente-militare-in-valoare-de-trei-milioane-de-dolari-in-urma-unui-acord-intre-cele-doua-tari-semnat-in-anul-2019.html>) (in Romanian). TVR Moldova.
27. "Exerciții demonstrative cu trageri de luptă la JCET-2023" (<https://news.ungheni.org/video-exercitii-demonstrative-cu-trageri-de-lupta-la-jcet-2023/>). *news.ungheni.org* (in Romanian). 2 November 2023.
28. "EDA - Morocco" (http://www.dsca.mil/sites/default/files/eda_public_report_2016-10-27.xls). Defense Security Cooperation Agency. Archived (https://web.archive.org/web/20170521023349/http://www.dsca.mil/sites/default/files/eda_public_report_2016-10-27.xls) from the original on 2017-05-21. Retrieved 2017-05-28.
29. "...: Ministerstwo Obrony Narodowej - serwis internetowy :: Galeria ::" (<https://web.archive.org/web/20101111002310/http://www.mon.gov.pl/pl/galeria/2374>). Archived from the original (<http://mon.gov.pl/pl/galeria/2374/>) on 2010-11-11. Retrieved 2014-03-25.
30. "NOVOS LANÇA GRANADAS AUTOMÁTICOS CHEGAM AO EXÉRCITO | Operacional" (<http://www.operacional.pt/novos-lanca-granadas-automaticos-chegam-ao-exercito/>) (in European Portuguese). Retrieved 2020-07-10.
31. Army (24 January 2012). *U.S. Army Weapons Systems 2013-2014* (<https://books.google.com/books?id=6bQYsnx7AH0C&q=M249+saudi&pg=PT869>). Skyhorse Publishing, Inc. ISBN 9781620879238. Archived (https://web.archive.org/web/20180318055104/https://books.google.com/books?id=6bQYsnx7AH0C&pg=PT869&lpg=PT869&dq=M249+saudi&source=bl&ots=vWYS5eB9Vy&sig=gJ9uDcSxNr5y0QD2hxmahl5RBXA&hl=en&sa=X&ved=0ahUK EwiJkZLTz_zYAhWGWxQKHU6sATIQ6AEIVTAJ#v=onepage&q=M249+saudi&f=false) from the original on 18 March 2018. Retrieved 17 March 2018 – via Google Books.
32. Janq Designs. "Special Operations.Com" (<https://web.archive.org/web/20120531143635/http://www.specialoperations.com/Foreign/Sweden/default.html>). Special Operations.Com. Archived from the original (<http://www.specialoperations.com/Foreign/Sweden/default.html>) on 2012-05-31. Retrieved 2012-11-24.
33. Svensk, Henrik. "Granatspruta 40mm Grsp" (<http://www.soldf.com/grsp.html>) (in Swedish). Soldf.com. Archived (<https://web.archive.org/web/20120624055440/http://www.soldf.com/grsp.html>) from the original on 2012-06-24. Retrieved 2012-11-24.
34. Ibp; USA (1 May 2007). *Thailand Army Weapon Systems Handbook* (<https://books.google.com/books?id=ddcRqnVDL9wC&pg=PA150>). Int'l Business Publications. p. 150. ISBN 978-1-4330-6196-7. Retrieved 8 August 2013.
35. "Bu Silah Kato'yu PKK'ya Dar Etti" (<https://web.archive.org/web/20120912091640/http://www.anahaberyorum.com/gundem/bu-silah-katoyu-pkky-a-dar-etti-h8892.html>). *anahaberyorum.com*. Archived from the original (<http://www.anahaberyorum.com/gundem/bu-silah-katoyu-pkky-a-dar-etti-h8892.html>) on 2012-09-12. Retrieved 5 November 2016.
36. "США передали Україні 40-мм автоматичні гранатомети Mk 19" (<https://mil.in.ua/uk/news/ssha-peredaly-ukrayini-40-mm-avtomatychni-granatomy-mk-19/>). *Мілітарний* (in Ukrainian). Retrieved 2022-02-03.

External links

- MK 19, 40-mm GRENADE MACHINE GUN, MOD 3 - Department of the Army, 2003 (https://armypubs.army.mil/doctrine/DR_pubs/dr_a/pdf/fm3_22x27.pdf)
 - General Dynamics (https://web.archive.org/web/20080409021726/http://www.gdatp.com/products/Gun_Systems/MK19/MK19.htm)
 - M430 40mm Cartridge High-explosive dual purpose (HEDP) round specs (<http://www.fas.org/man/dod-101/sys/land/m430.htm>)
-

Retrieved from "https://en.wikipedia.org/w/index.php?title=Mk_19_grenade_launcher&oldid=1289315517"